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Deep learning-based classification of wakefulness, sleep, and rumination states in dairy cows from polysomnography and RumiWatch data

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ABSTRACT

Sleep is essential for the health and welfare of dairy cattle, yet accurate monitoring remains challenging. While Polysomnography (PSG) serves as the gold standard for sleep assessment, its reliance on labor-intensive manual scoring of physiological signals constrains on-farm application, underscoring the need for accessible sensor systems such as the RumiWatch System (RWS). This study proposed DairySleepNet, a deep learning framework based on a selective state space model, for automated classification of wakefulness, sleep, and rumination states using PSG and RWS data, with the aim of supporting scalable annotations and on-farm deployment. The proposed framework adopted a 2-stage encoding architecture that effectively processed time-frequency representations converted from raw multichannel signals through the short-time Fourier transform, capturing both within-channel temporal features and cross-channel interactions. The method was evaluated using 77.13 h of usable PSG and RWS recordings from 7 cows, with approximately one 12-h segment retained for each cow. Animal-level cross-validation and a soft weighted sampling strategy were implemented to ensure robust evaluation. Experimental results demonstrated that DairySleepNet achieved a macro F1 of 86.37% and an accuracy of 90.80% when using PSG data, markedly outperforming both Transformer-based and machine learning baselines. These results supported the feasibility of automated sleep annotation from PSG recordings. Although combining PSG and RWS data reduced cross-fold variability, it did not improve the overall system performance. By contrast, classification based solely on RWS data remained unsatisfactory, underscoring the need for further methodological development before reliable on-farm sleep monitoring can be achieved. Future work should

prioritize fine-grained sleep staging, currently limited by scarce stage-specific labeled data, and broader data collection across diverse farm environments to enable robust and practical on-farm application.

Key words: animal welfare, sleep monitoring, rumination behavior, wearable sensor, multichannel signal classification

INTRODUCTION

Sleep is a fundamental biological process in mammals that supports neural restoration, immune function, and physiological homeostasis (Krueger et al., 2008; Cirelli and Tononi, 2008). In dairy cattle, sleep has received comparatively limited research attention, yet converging evidence suggests that it plays a meaningful role in health and welfare. The circadian timing system, which interacts closely with the sleep-wake cycle in dairy cattle, was shown to affect daily oscillations of metabolic hormones and core body temperature, and to prolong gestation length (Suarez-Trujillo et al., 2022). Although direct experimental evidence linking sleep deprivation to impaired health outcomes in cattle remains scarce, lying deprivation studies, which inherently restrict sleep opportunities, have demonstrated negative effects on immune markers, hormonal balance, and milk production (Kull et al., 2019). Sleep may therefore serve as a sensitive and integrative indicator of animal welfare in intensive farming systems (Cook, 2020), and accurate monitoring of sleep behavior is of considerable importance for welfare assessment and evidence-based management.

Measuring sleep in dairy cattle nevertheless remains challenging. Visual observation and video-based behavioral scoring are labor-intensive and subject to observer bias, and behavioral proxies such as lying posture substantially overestimate total sleep time in adult cows compared with electrophysiological measurements (Ternman et al., 2014). Polysomnography (PSG), which combines electroencephalography (EEG), electromyography (EMG), and electrooculography (EOG), is con-

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sidered the gold standard for sleep assessment in both humans and animals (Guillodo et al., 2020; Ternman et al., 2012). Manual scoring of PSG recordings by trained experts is, however, time-consuming and impractical for large-scale or long-term monitoring in farm settings. These limitations highlight the need for automated approaches to sleep classification in cattle.

In human sleep research, automated sleep stage classification has been studied extensively. Traditional machine learning approaches such as support vector machines and random forests achieved moderate success when combined with time-domain features (Aboalayon et al., 2016), whereas deep learning methods have demonstrated superior performance by automatically learning hierarchical representations from raw signals or time-frequency spectrograms (Supratak et al., 2017; Palotti et al., 2019; Zhou et al., 2022). The Transformer architecture is able to model long-range dependencies through self-attention (Vaswani et al., 2017; Eldele et al., 2021), although its quadratic computational complexity with respect to sequence length poses scalability challenges when multi-channel physiological recordings are processed. Mamba, a selective state space model proposed by Gu and Dao (2023), achieves linear-time complexity through input-dependent state transitions and offers comparable modeling capacity at substantially lower computational cost. The selective mechanism enables the model to adaptively emphasize informative time steps while suppressing irrelevant segments, a property that is particularly suited to physiological signals in which transient events coexist with sustained background activity. Architectures of this type have been applied successfully to EEG-based tasks in humans (Li et al., 2025; Guo et al., 2025; Lee et al., 2025), but their application to livestock sleep classification has not yet been explored.

In contrast to the extensive research on human sleep, automated sleep classification in cattle remains underexplored. Existing studies on automated cattle behavior classification have relied mainly on accelerometer data and traditional machine learning algorithms with time-domain features, and have addressed activity recognition rather than sleep (Riaboff et al., 2020; Riaboff et al., 2022). Where sleep has been considered, PSG has been employed exclusively as a labeling reference, while the classification models themselves operate on derived physiological or behavioral features such as heart rate and neck muscle activity (Walch et al., 2019; Hunter et al., 2021). To our knowledge, no study has used PSG signals directly as input to machine learning or deep learning models for cattle sleep classification, and deep learning techniques that leverage time-frequency representations of raw multi-channel recordings have not been systematically investigated. A further limitation is that rumination has generally been excluded rather than mod-

eled, even though it is a predominant behavior in cattle and introduces electrophysiological artifacts that overlap with sleep-related signal patterns.

Several additional challenges remain in this domain. From a signal modeling perspective, EEG recordings contain rich frequency-specific information that traditional time-domain features may fail to capture (Chambon et al., 2018), and multi-channel PSG recordings require effective integration of cross-channel information while preserving channel-specific characteristics. Furthermore, sleep epochs are considerably less frequent than wake and rumination epochs in typical recordings, which complicates model training and evaluation. The feasibility of using accessible on-farm sensor data, rather than research-grade PSG equipment, for predicting sleep states has not been systematically assessed, although this question is central to translating sleep monitoring into routine welfare assessment. In addition, many existing studies on cattle behavior classification employed sample-level cross-validation, in which epochs from the same animal may appear in both training and testing sets, leading to optimistic performance estimates that do not reflect generalization to unseen individuals.

The objective of this study was to develop and evaluate an automated approach for classifying wakefulness, sleep, and rumination states in dairy cows, with the aim of reducing the labor associated with manual PSG scoring and assessing the feasibility of on-farm sleep monitoring. To this end, we introduce DairySleepNet, a deep learning framework based on a selective state space model, and apply it to recordings obtained from PSG, from the RumiWatch System (RWS), and from their combination. Through animal-level cross-validation on a data set of 7 dairy cows, we further examine whether wearable rumination and activity sensors can approach the classification performance achieved with full electrophysiological recordings.

MATERIALS AND METHODS

Animals and Experimental Setup

All procedures were approved by the Cantonal Veterinary Office of Zürich (approval number: ZH042/2023) and conducted at the AgroVet-Strickhof Dairy Farm in Lindau, Switzerland. Nine multiparous Holstein cows with calm temperament were selected from the control group of a previous study at the same facility. All animals were clinically healthy and habituated to human contact, facilitating instrumentation setup. Cows were housed in a tie-stall barn with individual feeding and watering facilities. A 48-h acclimation period was provided before data collection. Because the recorder required a battery change every 12 h, each animal was scheduled for 2 consecutive

12-h sessions, one covering daytime (08:00–20:00 h) and one covering nighttime (20:00–08:00⁺¹ h), so that sleep behavior was captured across different periods of the circadian cycle. Among the recordings obtained, the proportion of scored sleep epochs varied considerably between sessions, and only sessions in which sleep accounted for more than 10% of the scored epochs were retained to ensure sufficient representation of the minority state. This selection yielded one 12-h session from each of 7 cows, comprising 3 daytime sessions and 4 nighttime sessions. Natural lighting was maintained throughout recordings.

Data Collection

The PSG recordings were obtained using the non-invasive methodology adapted from Ternman et al. (2012). Approximately 24 h before recording, the electrode attachment sites on the forehead, periorbital regions, and neck were shaved. The sites were cleaned with 70% ethanol approximately 3 h before recording and allowed to dry. A 10-electrode montage was applied using disposable pre-gelled snap electrodes, comprising 4 EEG electrodes, 2 EOG electrodes, 2 EMG electrodes, 1 reference electrode, and 1 ground electrode, as shown in Figure 1. The EEG electrodes were arranged in a 2×2 configuration on the forehead between the eye level and the cranial aspect of the horn base. The reference electrode was positioned at the center of the EEG array, and the ground electrode was positioned caudal to the horn base. The EOG electrodes were positioned above the left eye and below the right eye, whereas the EMG electrodes were positioned bilaterally over the cervical portion of the trapezius muscle. The electrodes were secured using Ergo cyanoacrylate adhesive, and the leads were bundled in a protective cable holder to reduce tension. The recorder and a counterweight were secured to an udder holder positioned over the dorsal region. Signals were recorded at 500 Hz with 24-bit resolution using an Embla ST+ Proxy connected to an Embletta MPR PG recorder (Natus Medical Inc., Middleton, WI, USA). Electrode impedance was checked before recording and after any electrode replacement or reconnection. Values between 0.2 and 5 k Ω were considered acceptable. Electrode status was evaluated at 30-min intervals, and waveform quality was rechecked after reconnection. A maximum of 2 cows were recorded per session. Respiratory movements were simultaneously recorded at 30 Hz using an XactTrace belt positioned around the abdomen. The resulting PSG data set comprised 4 EEG, 2 EOG, 2 EMG, and 1 respiratory, in total, 9 channels.

The RumiWatch system (RWS; Itin and Hoch GmbH, Liestal, Switzerland) was deployed simultaneously with PSG recordings. The RWS is a noseband-mounted sensor comprising a glycol-filled pressure tube and a triaxial

accelerometer (Rombach et al., 2018). Sensor data were logged at 10 Hz via an integrated data logger and stored on a secure digital memory card. Raw data were extracted using RumiWatch Manager software (v2.1.0.0) and exported in CSV format for subsequent analysis without processing by RumiWatch Converter. The RWS provided 4 channels: triaxial acceleration (X, Y, Z) and pressure.

Data Annotation

In human sleep research, polysomnographic signals are typically segmented into 30-s epochs and classified according to standardized criteria (Rechtschaffen and Kales, 1968; Korkalainen et al., 2021). For cattle, Ternman et al. (2012) proposed a 4-stage classification comprising wake, drowsing, non-rapid eye movement

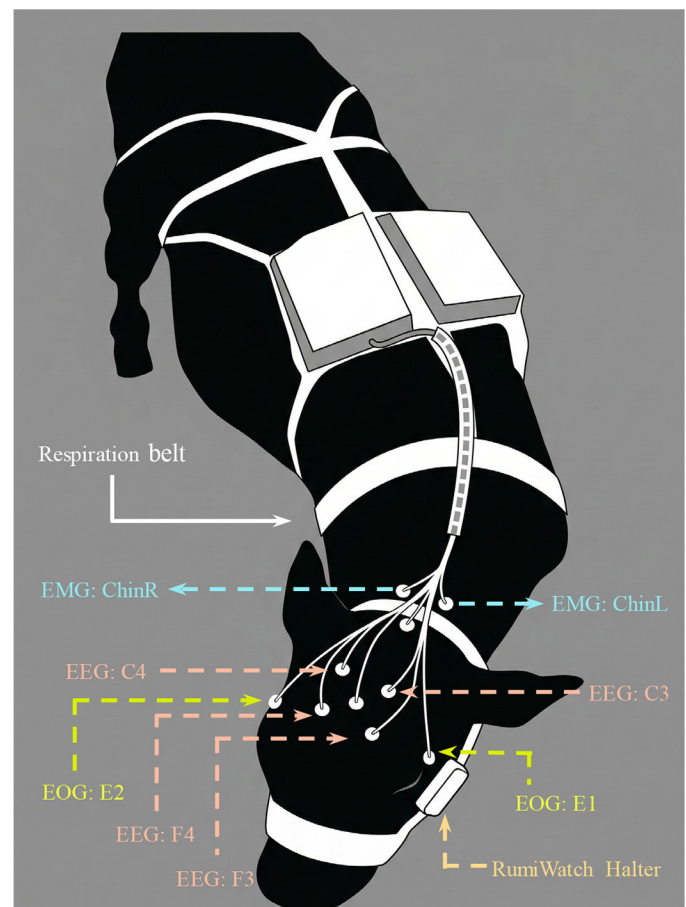


Figure 1. Schematic of the polysomnography (PSG) electrode montage and RumiWatch system (RWS) placement on dairy cows. The PSG montage consists of 4 EEG electrodes positioned on the forehead, 2 EOG electrodes placed periorbitally, 2 EMG electrodes attached bilaterally on the trapezius muscle, one ground electrode, and one reference electrode. The RWS noseband sensor provides triaxial acceleration and pressure channels. EEG, electroencephalography; EMG, electromyography; EOG, electrooculography.

(NREM) sleep, and rapid eye movement (REM) sleep. Based on that, in the present study, PSG recordings were manually scored in 30-s epochs by a well-trained annotator using RemLogic v3.4.3 software, based on synchronized EEG, EOG, and EMG signals. Given that rumination is a predominant behavior in cattle and generates distinct physiological signatures that differ markedly from both wake and sleep, it was included as a separate vigilance state. Five states were therefore defined: Wake (W), Rumination (R), Drowsing (N1), NREM sleep (N2), and REM sleep (N3).

Wake was characterized by desynchronized EEG with predominant high-frequency activity, pronounced neck EMG tone, and irregular eye movements. Cows were either standing or lying with the head supported by active neck musculature. Rumination was identified by highly rhythmic spike-wave artifacts in EEG, EMG, and EOG traces caused by masticatory muscle contractions. This pattern occurred in approximately one-minute cycles, interrupted briefly during swallowing and regurgitation. Classification relied primarily on rhythmic EMG/EOG patterns combined with behavioral confirmation.

Drowsing represented a transitional state featuring partially synchronized EEG with recurring K-complexes and reduced but present EMG activity. Sleep of the NREM stage was characterized by synchronized low-frequency EEG dominated by slow waves and further reduced muscle tone. In contrast, REM sleep exhibited desynchronized EEG resembling wakefulness, marked muscle atonia, and characteristic rapid eye movement bursts. Behaviorally, cows rested with their heads unsupported by neck musculature. When multiple states occurred within a single epoch, the predominant state (>50% duration) was assigned. Epochs were marked as missing data when all channels of any one signal modality (EEG, EMG, or EOG) exhibited complete signal loss, whereas partial channel loss within a modality did not lead to exclusion, since the remaining channels still retained related information from that modality.

Because the primary objective of this study is the automated prediction of broad sleep behavior rather than the detailed differentiation of sleep stages, the stages of Drowsing, NREM sleep, and REM sleep were consolidated into a single Sleep (S) class for model training. Consequently, the final classification task comprised 3 categories. The proportion of data for each category in the data set was 45.59% for the Wake class, 41.62% for the Rumination class, and 12.79% for the Sleep class. The state labels derived from PSG were assigned to the corresponding RWS epochs matched by timestamps. The summary of the final experimental data set is as shown in Table 1.

Signal Preprocessing

Epochs marked as missing data were excluded before preprocessing. The remaining data were processed using 2 distinct strategies for the proposed deep learning model and the baseline machine learning models, respectively.

For deep learning models, to ensure temporal consistency across multimodal signals, all recordings were resampled to a unified sampling rate of 100 Hz using the MNE-Python library. Time-frequency representations were then extracted using the short-time Fourier transform (STFT; Allen and Rabiner, 1977). A 2-s Hanning window with 50% overlap was applied to each epoch. The FFT length was set to 256 points. Due to the conjugate symmetry of the FFT for real-valued signals, only the positive frequency components were retained, resulting in 128 frequency bins. According to Nyquist Theorem (Shannon, 1949), this corresponds to a frequency range of 0–50 Hz, which encompasses the physiologically relevant bands for sleep analysis, including delta (0.5–4 Hz), theta (4–8 Hz), α (8–13 Hz), and β (13–30 Hz) rhythms. Spectrograms were temporally standardized to 29 frames per epoch, yielding a tensor of shape (29 × 128) per channel as shown in Figure 2. A logarithmic transformation was then applied to compress the dynamic range. For baseline machine learning models, 6 time-domain

Table 1. Summary of the experimental data set. Statistics showing the distribution of behavioral epochs (Wake, Sleep, and Rumination), the total number of epochs, and the overall monitoring duration per animal across the seven dairy cows

Cow ID	Behavior classes			Total Epochs	Duration (hour)
	Wake	Sleep	Rumination		
Cow 1	766	158	234	1,158	9.65
Cow 2	473	139	780	1,392	11.60
Cow 3	346	150	672	1,168	9.73
Cow 4	853	222	315	1,390	11.58
Cow 5	756	201	451	1,408	11.73
Cow 6	449	142	750	1,341	11.18
Cow 7	577	172	650	1,399	11.66

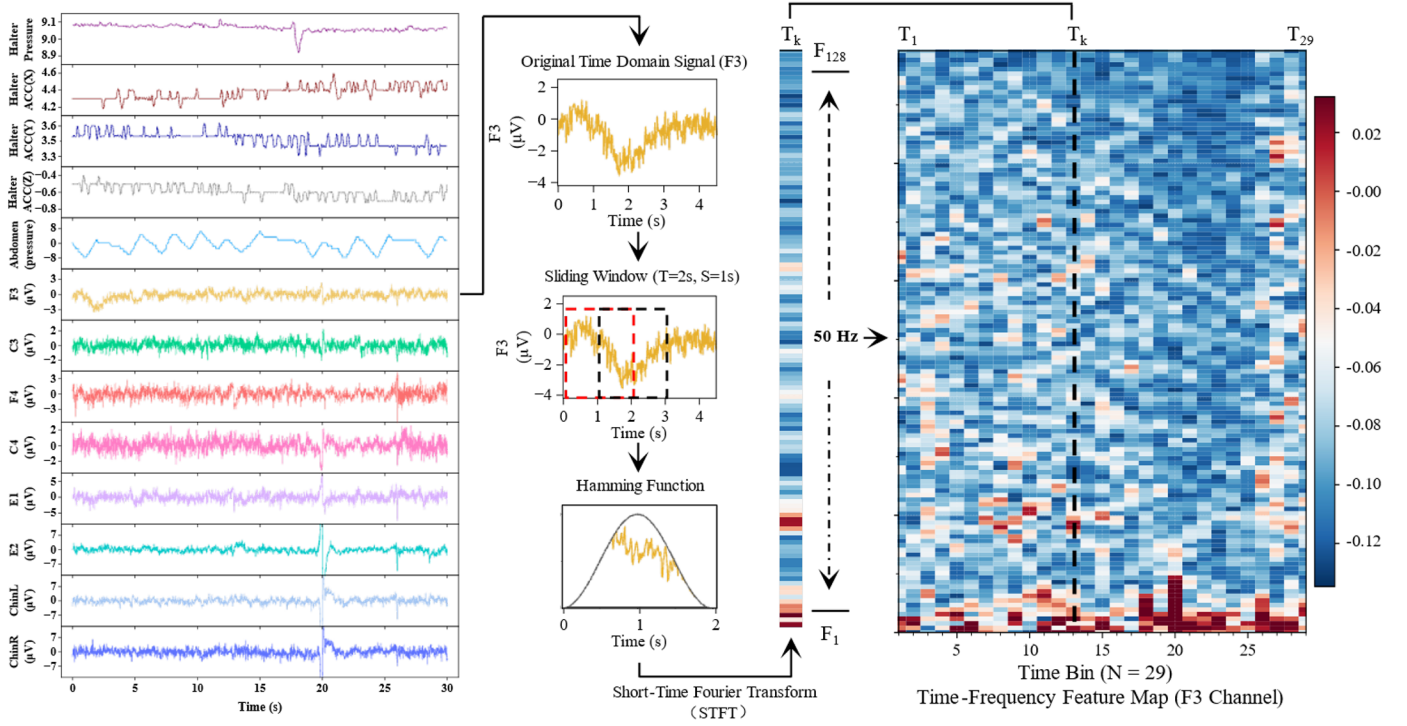


Figure 2. Illustration of the short-time Fourier transform (STFT) preprocessing pipeline. A representative 30-s Sleep epoch (Cow 4, epoch 40) is shown. Raw signals are resampled to 100 Hz and transformed using a 2-s Hanning window with 50% overlap and 256-point FFT, yielding a time-frequency spectrogram of 29 temporal frames and 128 frequency bins (0–50 Hz) per channel.

statistical features were extracted from each channel per epoch: mean, standard deviation, root mean square, peak-to-peak amplitude, skewness, and kurtosis. This yielded a 54-dimensional feature vector from the 9 PSG channels and a 24-dimensional vector from the 4 RWS channels. Additionally, signal magnitude area was computed from the 3 acceleration axes to quantify overall movement intensity. The concatenated feature vector comprised 79 dimensions per epoch. For both representations, features calculated from each channel were normalized to zero mean and unit variance across all epochs and served as the direct input to different models.

Model Architecture

DairySleepNet maps multichannel time-frequency representations to vigilance states through a 2-stage encoding architecture, as shown in Figure 3. Each channel is first processed independently to preserve channel-specific temporal patterns. The resulting representations are then fused to capture interactions across channels. The input is represented as a tensor $\mathcal{X} \in \mathbb{R}^{C \times T \times F}$, where $0 < C \leq 13$ denotes the number of physiological channels, $T = 29$ denotes the number of temporal frames, and $F = 128$ denotes the number of frequency bins.

Sequence modeling in both encoding stages is performed using Mamba blocks based on a selective state space model (Gu and Dao, 2023). Within each epoch, the model propagates a latent state $h(t) \in \mathbb{R}^N$ across temporal frames according to:

$$h'(t) = \bar{A}_t h(t) + \bar{B}_t x(t), \quad y(t) = C_t h(t), \quad [1]$$

where $x(t)$ and $y(t)$ denote the input and output at temporal frame t , respectively. The discrete parameters $\bar{A}_t$ and $\bar{B}_t$ are obtained from the corresponding continuous-time parameters through zero-order-hold discretization. In a conventional state space model, the state update parameters remain fixed across time. In the selective formulation, $\bar{B}_t$, C_t , and the discretization interval Δ_t depend on the current input:

$$\begin{aligned} B_t &= \text{Linear}_B(x_t), \\ C_t &= \text{Linear}_C(x_t), \\ \Delta_t &= \text{softplus}[\text{Linear}_\Delta(x_t)]. \end{aligned} \quad [2]$$

This input-dependent parameterization regulates the amount of new information incorporated into the latent state and the amount of previous information retained

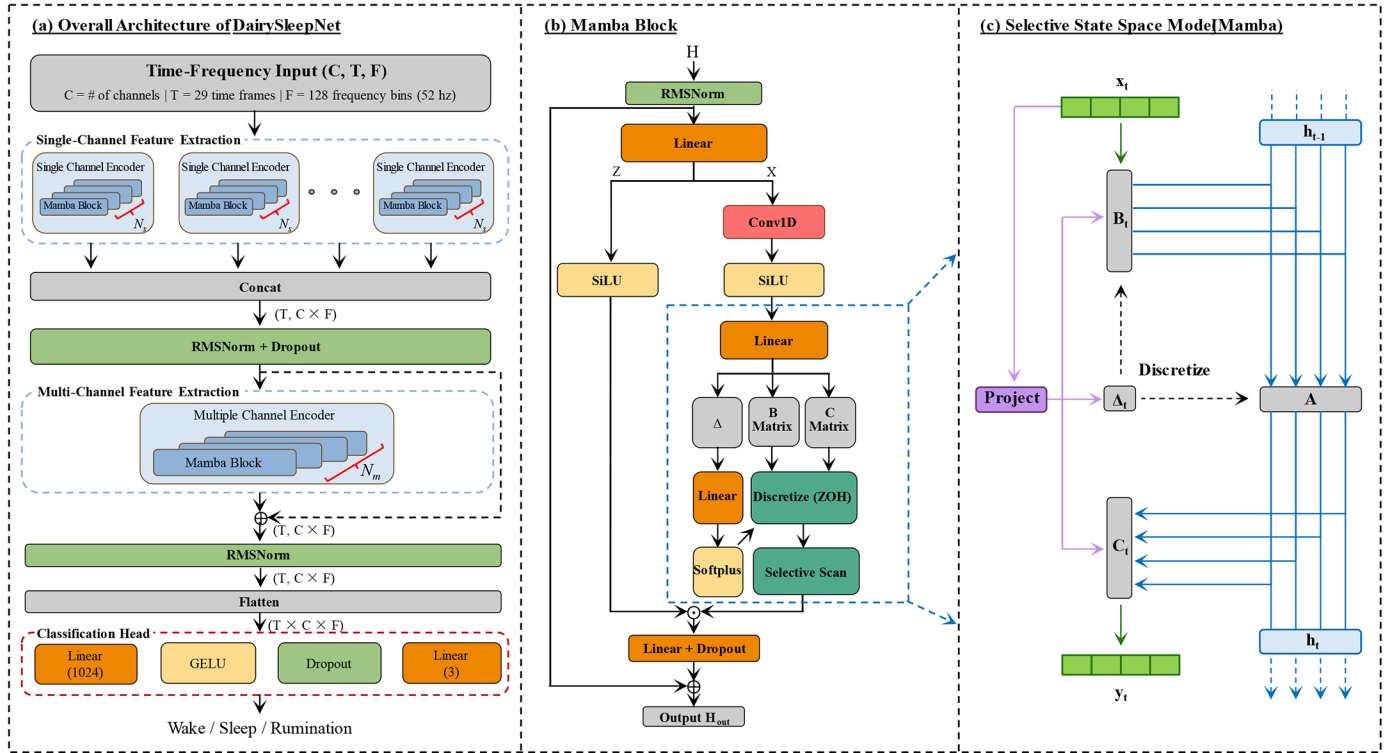


Figure 3. Architecture of DairySleepNet. (a) Overall pipeline showing the 2-stage encoding strategy. Multichannel time-frequency spectrograms are processed by parallel single-channel Mamba encoders, concatenated along the feature dimension, and passed through a multichannel fusion encoder with residual connections before classification. (b) Internal structure of a single Mamba Block. (c) Selection mechanism within Mamba block.

at each temporal frame. It may therefore facilitate the identification of temporally localized patterns, such as K-complexes, within periods of relatively stable background activity. Each Mamba block combines the selective recurrence with a depthwise convolution for local feature extraction and a multiplicative gate for output regulation. The block preserves the shape of the input representation.

As shown in Figure 3(a), in the first encoding stage, N_S stacked Mamba blocks are applied independently to each channel. The frequency bins form the feature dimension, whereas the temporal frames form the sequence dimension. The channel-level outputs are concatenated along the feature dimension and regularized using dropout (Srivastava et al., 2014) and layer normalization (Ba et al., 2016). The resulting representation is processed by a fusion encoder containing N_M stacked Mamba blocks and a residual connection (He et al., 2016):

$$M_{fused} = Encoder_{fusion}(M_{cat}) + M_{cat}. \quad [3]$$

The temporal recurrence has linear computational complexity with respect to T . Concatenation increases the feature dimension with the number of channels without

increasing the temporal sequence length. Finally, the fused representation is flattened and mapped to the 3 target classes through 2 fully connected layers with GELU activation (Hendrycks and Gimpel, 2016) and dropout.

Baseline Models

To contextualize the performance of DairySleepNet, we compare it against 2 categories of baselines: 4 conventional machine learning classifiers operating on the time-domain features, and MultiChannelSleepNet, a Transformer-based deep learning model operating on the same time-frequency input as the proposed model.

Four machine learning classifiers were Random Forest (RF), Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and Logistic Regression (LR). These classifiers are widely adopted in livestock behavior classification studies and represent a range of modeling paradigms from ensemble methods to linear models (Breiman, 2001; Cortes and Vapnik, 1995; Rumelhart et al., 1986). All 4 were configured with class-balanced weighting to mitigate the influence of the imbalanced class distribution in the data set.

For the deep learning baseline, we adopted MultiChannelSleepNet (Dai et al., 2023), a Transformer-based

model that has demonstrated strong performance in multichannel sleep stage classification on public human PSG data sets (Kemp et al., 2000; Goldberger et al., 2000; Quan et al., 1997; Zhang et al., 2018). MultiChannelSleepNet employs Transformer encoders with multi-head self-attention for both single-channel feature extraction and multichannel feature fusion and shares a comparable hierarchical encoding structure with the proposed model. Including this baseline enables a direct comparison between attention-based and state-space-based sequence modeling paradigms under a similar architectural framework. To ensure a fair evaluation, MultiChannelSleepNet was trained with the same time-frequency input representation, cross-validation scheme, and class-balancing strategy as DairySleepNet.

Training Strategy

To evaluate generalization at the animal level, we adopted a subject-level K-fold cross-validation scheme with K set to 4. The 7 cows were randomly partitioned into 4 non-overlapping groups, with each group serving as the held-out test set in a fold, so that no data from any test animal appeared in the corresponding training set. Within each fold, 10% of the training data was held out as a stratified validation set for early stopping. The same partitions were applied consistently across all models.

To address the substantial class imbalance, a soft weighted sampling strategy was employed during mini-batch construction for the deep learning models. Per-class sampling weights were computed as the inverse square root of the class frequency:

$$w_k = \frac{1}{\sqrt{n_k}}, \quad [4]$$

where n_k denotes the number of training samples in class k . Compared with standard inverse-frequency weighting, the square-root formulation provides moderate rebalancing that increases minority-class representation while reducing the risk of overfitting to repeated examples. For the machine learning baselines, an equivalent effect was obtained through the built-in class-weight parameter of each algorithm, which adjusts the loss contribution of each class in inverse proportion to its frequency.

Parameters

For the deep learning models, the AdamW optimizer (Loshchilov and Hutter, 2019) was adopted with a learning rate of 5×10^{-6} and a weight decay of 10^{-2} . The training objective was the cross-entropy loss with label smoothing of 0.1, gradient norms were clipped to 1.0,

and the mini-batch size was 32. Training ran for a maximum of 200 epochs with an early stopping patience of 20 epochs monitored on the validation macro F1, and the checkpoint with the highest validation macro F1 was retained for evaluation. All experiments were implemented in PyTorch 2.5.1 and scikit-learn 1.5.2 and executed on a cluster of 8 NVIDIA RTX 4090 GPUs.

In DairySleepNet, the single-channel encoders used $N_S = 4$ stacked Mamba Blocks and the multichannel fusion encoder used $N_M = 4$, with an SSM state dimension was set to 16, a convolutional kernel size to 4, and an inner dimension expansion factor to 2. A dropout rate of 0.2 was applied throughout the network. In MultiChannelSleepNet, 8 Transformer encoders with 8 attention heads were used for single-channel feature extraction, and 4 for multichannel fusion, with dropout rates of 0.1 and 0.5 for the encoder sub-layers and the fusion block respectively, following the original study (Dai et al., 2023). Both models used a feature dimension of $D = 128$, matching the number of frequency bins in the input representation, and a classification head with 1024 hidden units.

For the machine learning baselines, RF was configured with 300 trees, a minimum leaf size of 5, and no maximum depth constraint. The SVM used a radial basis function kernel with a regularization parameter $C = 10$. The MLP employed 3 hidden layers of 256, 128, and 64 units with an adaptive learning rate, L2 regularization ($\alpha = 0.001$), and early stopping with a patience of 20 iterations. The LR used L2 regularization with $C = 1.0$ and the L-BFGS solver.

Evaluation Metrics

Model performance was assessed using both overall and per-class metrics (Lewis et al., 1996; Cohen, 1960). For per-class evaluation, we reported precision (PR), recall (RE), and F1-score (F1) for each of the 3 target classes, denoted by the index k representing Wake, Sleep and Rumination. These metrics are defined as:

$$\begin{aligned} PR_k &= \frac{TP_k}{TP_k + FP_k}, \\ RE_k &= \frac{TP_k}{TP_k + FN_k}, \\ F1_k &= \frac{2 \cdot PR_k \cdot RE_k}{PR_k + RE_k} \end{aligned} \quad [5]$$

where TP_k , FP_k , and FN_k denote the number of true positives, false positives, and false negatives for class k , respectively. Precision measures the proportion of predicted instances for a given class that are correctly

classified, while recall measures the proportion of actual instances that are correctly identified. The F1-score is the harmonic mean of the 2, balancing both aspects into a single metric. These per-class metrics are especially relevant in this study, as the ability to correctly identify the minority Sleep class is of primary interest for sleep monitoring applications.

For overall evaluation, we reported accuracy (ACC), macro-averaged F1-score (Macro F1), and Cohen's kappa (κ). Accuracy is the proportion of correctly classified epochs across all classes:

$$\text{ACC} = \frac{\sum_{k=1}^K TP_k}{N}, \quad [6]$$

where K is the number of classes and N is the total number of test samples. Macro F1 is computed as the unweighted arithmetic mean of the per-class F1-scores:

$$\text{Macro F1} = \frac{1}{K} \sum_{k=1}^K F1_k. \quad [7]$$

This metric treats all classes equally regardless of their sample size, making it particularly informative under class-imbalanced conditions. Cohen's kappa quantifies the agreement between predicted and true labels while correcting for chance agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad [8]$$

where p_o denotes the observed agreement (equivalent to accuracy) and p_e denotes the expected agreement under random prediction. This metric provides a more conservative assessment than accuracy alone, as a high accuracy can be achieved by simply predicting the majority class. All metrics were computed on the held-out test set of each fold, and results are reported as the mean and standard deviation across the 4 folds.

RESULTS AND DISCUSSION

Overall Classification Performance

Figure 4 presents the training loss and validation macro F1 as a function of training epoch for a representative fold (Fold 4) of DairySleepNet with PSG input. The training loss decreased steadily over the first 30 epochs and converges thereafter, while the validation macro F1 rose rapidly during the initial epochs and stabilized around epoch 40, reaching a peak value of 0.92. No substantial

divergence was observed between the training and validation curves, indicating that the model did not exhibit significant overfitting under the adopted regularization strategy. Similar convergence behavior was observed across the remaining folds and modalities.

Table 2 presents the summed confusion matrices and per-class evaluation metrics for DairySleepNet under the 3 input modalities. With PSG input, DairySleepNet achieved an overall accuracy of 90.77%, a macro F1 of 86.64%, and a Cohen's kappa of 0.85. It performed strongly on Wake (F1 = 90.14%) and Rumination (F1 = 96.88%), while Sleep, the minority class, obtained an F1 of 72.90% with a precision of 75.54% and a recall of 70.44%. The confusion matrix revealed that the most frequent misclassification occurred between Wake and Sleep, with 29.56% Sleep epochs misclassified as Wake and 6.33% Wake epochs misclassified as Sleep. This pattern is consistent with the physiological similarity between quiet wakefulness and light sleep states, where EEG characteristics partially overlap. The strong separation of Rumination reflects the highly discriminative nature of the rhythmic masticatory artifacts that appear in the EEG, EMG, and EOG traces, which after transformation by the STFT produce spectral signatures that differ markedly from those of both wake and sleep.

With RWS input, performance decreased substantially across all metrics, with an accuracy of 74.63%, a macro F1 of 71.13%, and a kappa of 0.60. The Sleep class showed a markedly different error profile, in which recall increased to 73.48% but precision dropped to 46.06%, giving an F1 of 56.62%. The confusion matrix showed that 22.01% of Wake epochs were incorrectly classified as Sleep, which accounts for the low precision. This behavior indicates that accelerometer and pressure signals, which capture only mechanical aspects of head movement and jaw activity, lack the spectral resolution needed to separate behaviorally quiescent states such as quiet wakefulness and light sleep. A nonzero confusion between Sleep and Rumination also emerged under this modality but was virtually absent under PSG, which confirms that the reliable separation of all 3 states depends on the spectral information contained in the neurophysiological signals.

With the combined PSG+RWS input, the model achieved an accuracy of 90.04%, a macro F1 of 86.30%, and a kappa of 0.84. Compared with PSG-only, all 3 overall metrics decreased slightly, while the Sleep F1 improved marginally from 72.90% to 73.33% through an increase in recall at the expense of precision. Despite this small decrease in mean performance, the cross-fold variability was notably lower for the combined modality, with a macro F1 standard deviation of 2.67% compared with 4.48% for PSG alone. This suggests that the additional RWS channels contributed to more stable predictions

Table 2. Cross-validation classification performance of DairySleepNet using polysomnography (PSG), RumiWatch System (RWS) and both modalities (PSG+RWS), with summed confusion matrices and per-class evaluation metrics aggregated across all folds

Modality		Predicted			DairySleepNet					
		W	S	R	PR	RE	F1	Macro F1	ACC	κ
PSG	W	3892	267	61	88.15	92.23	90.14	86.64	90.77	0.85
	S	350	834	0	75.54	70.44	72.90	(86.37 $\pm$ 4.48)	(90.80 $\pm$ 2.31)	(0.85 $\pm$ 0.04)
	R	173	3	3676	98.37	95.43	96.88			
RWS	W	2863	929	428	77.19	67.84	72.22	71.13	74.63	0.60
	S	259	870	55	46.06	73.48	56.62	(71.88 $\pm$ 4.66)	(75.70 $\pm$ 5.12)	(0.61 $\pm$ 0.08)
	R	587	90	3175	86.80	82.42	84.55			
PSG+RWS	W	3773	367	80	88.86	89.41	89.13	86.30	90.04	0.84
	S	285	899	0	70.90	75.93	73.33	(86.09 $\pm$ 2.67)	(89.86 $\pm$ 2.01)	(0.83 $\pm$ 0.03)
	R	188	2	3662	97.86	95.07	96.44			

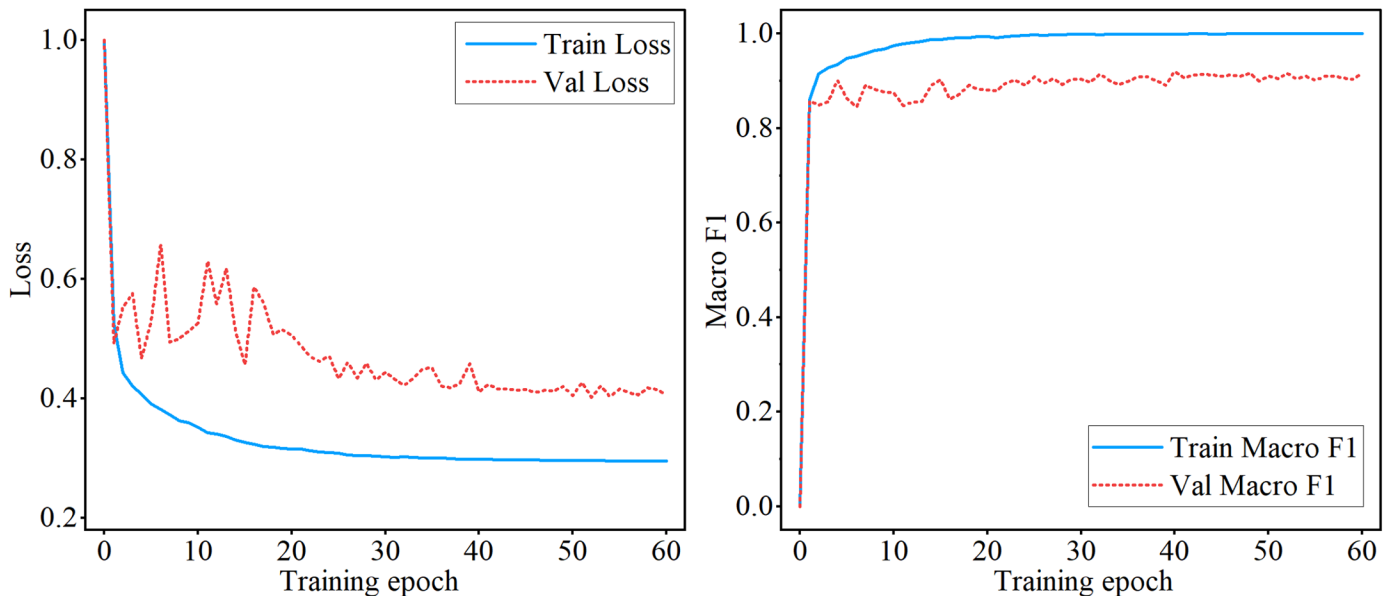
across different animals, most likely through the direct mechanical measurement of jaw movement provided by the pressure sensor, even though they added little discriminative information beyond that already encoded in the PSG representation.

Taken together, these results indicate that the neurophysiological signals provided by PSG offer the most informative representation for sleep classification in dairy cows. The RWS data alone were insufficient for reliable sleep detection, as indicated by the low precision and the high confusion between Wake and Sleep. This has a direct bearing on deployment. The noseband sensor is inexpensive and already used on commercial farms for feeding and rumination monitoring, which makes it an attractive basis for scalable welfare assessment, but its accuracy for sleep detection does not yet allow it to replace PSG, and PSG in turn depends on research-grade equipment

and expert scoring that are impractical for routine herd-level use. A viable on-farm system will therefore require richer information from the noseband signals or the addition of complementary low-cost measures, rather than direct transfer of the PSG-based pipeline evaluated here. The combination of the 2 modalities did not improve mean performance relative to PSG alone but reduced cross-fold variability, which may be advantageous in longitudinal monitoring where prediction stability across individuals is prioritized.

Comparison with Baseline Models

Table 3 compares the classification performance of DairySleepNet against 5 baseline models across the 3 input modalities, and Figure 5 presents the corresponding normalized confusion matrices. With PSG input,

**Figure 4.** Training curves of DairySleepNet with PSG input for a random fold (Fold 4). The model converges within approximately 40 epochs, with early stopping triggered at the point of highest validation macro F1.

DairySleepNet achieved the highest performance among all models, with a macro F1 of 86.37%, an accuracy of 90.80%, and a kappa of 0.85. MultiChannelSleepNet obtained comparable results (macro F1 = 84.32%), while the best machine learning baseline, RF, reached a macro F1 of 77.27%. The remaining machine learning classifiers achieved macro F1 ranging from 67.66% (LR) to 75.24% (SVM). The 2 deep learning models, therefore, improved the macro F1 by 9.10 and 7.05 percentage points over RF.

This advantage reflects the informativeness of the time-frequency representation together with the capacity of deep architectures to learn hierarchical feature abstractions. Time-domain features summarize the amplitude distribution within each epoch but discard temporal ordering and frequency composition, whereas the spectrograms obtained by the STFT preserve both, which enables the models to distinguish states that share similar amplitude statistics but differ in frequency content, such as quiet wakefulness and light sleep. The advantage was most pronounced for Rumination, where both deep learning models achieved a recall of 0.95 compared with 0.76 to 0.82 for the machine learning baselines, indicating that time-frequency representations are particularly effective at capturing the spectral signatures of rumination activity.

The performance hierarchy changed under RWS input. Among all models, RF achieved the highest macro F1 of 74.03%, exceeding both deep learning models, and the SVM exhibited the highest cross-fold variability with a macro F1 standard deviation of 9.21%. This reversal indicates that the benefit of the time-frequency representation depends on the spectral richness of the input signal. Accelerometer data are dominated by low-frequency

movement patterns for which compact time-domain statistics capture sufficient discriminative information, so the additional representational capacity of the deep learning models offers little advantage and may even reduce stability across animals.

With the combined PSG and RWS input, DairySleepNet again achieved the best performance (macro F1 = 86.09%), followed by MultiChannelSleepNet and RF. The benefit of modality fusion varied considerably across models. Both RF and LR improved substantially when the RWS channels were added to the PSG input, with gains in macro F1 of 4.13 and 6.51 percentage points, whereas both deep learning models showed marginal decreases. This divergence suggests that deep learning models already operate near the information ceiling of the PSG representation, leaving limited room for the additional channels to contribute, while the lower-dimensional feature space used by RF and LR retains capacity to benefit from supplementary modalities. The fusion narrowed the gap between RF and DairySleepNet from 9.10 percentage points under PSG alone to 4.69 percentage points.

Because DairySleepNet and MultiChannelSleepNet share an identical architectural framework and differ only in the sequence modeling backbone, their comparison provides a controlled evaluation of the selective state space model against the Transformer for multi-channel physiological signal classification. DairySleepNet outperformed MultiChannelSleepNet across all modalities, most clearly under PSG (macro F1 = 86.37% vs. 84.32%, accuracy = 90.80% vs. 87.76%). Two properties of the selective state space model are relevant to this task. The input-dependent parameterization allows the model to emphasize informative time-frequency frames, such as those containing transient sleep events, while attenuat-

Table 3. Comparison of classification performance between DairySleepNet and baseline models across the three input modalities, reported as mean and standard deviation of accuracy (ACC), macro F1, and Cohen's kappa (κ) over all cross-validation folds

Methods	Modality	Macro F1	ACC	κ
RF	PSG	77.27 $\pm$ 4.57	79.95 $\pm$ 4.63	0.67 $\pm$ 0.08
	RWS	74.03 $\pm$ 2.81	79.11 $\pm$ 2.61	0.65 $\pm$ 0.04
	PSG+RWS	81.40 $\pm$ 2.68	85.61 $\pm$ 0.99	0.76 $\pm$ 0.02
SVM	PSG	75.24 $\pm$ 5.19	78.07 $\pm$ 4.91	0.64 $\pm$ 0.09
	RWS	65.08 $\pm$ 9.21	68.19 $\pm$ 10.26	0.49 $\pm$ 0.16
	PSG+RWS	74.02 $\pm$ 5.47	76.53 $\pm$ 5.20	0.61 $\pm$ 0.09
MLP	PSG	69.86 $\pm$ 7.39	75.44 $\pm$ 6.71	0.59 $\pm$ 0.12
	RWS	67.39 $\pm$ 3.86	72.16 $\pm$ 2.84	0.54 $\pm$ 0.05
	PSG+RWS	70.20 $\pm$ 4.77	74.49 $\pm$ 4.19	0.57 $\pm$ 0.07
LR	PSG	67.66 $\pm$ 7.42	70.11 $\pm$ 8.18	0.53 $\pm$ 0.12
	RWS	66.01 $\pm$ 1.59	69.37 $\pm$ 2.51	0.52 $\pm$ 0.04
	PSG+RWS	74.17 $\pm$ 4.19	76.66 $\pm$ 5.36	0.62 $\pm$ 0.09
MultiChannelSleepNet	PSG	84.32 $\pm$ 3.42	87.76 $\pm$ 3.39	0.80 $\pm$ 0.05
	RWS	68.02 $\pm$ 2.50	72.65 $\pm$ 2.83	0.56 $\pm$ 0.05
	PSG+RWS	83.40 $\pm$ 2.67	87.34 $\pm$ 2.57	0.79 $\pm$ 0.04
DairySleepNet	PSG	86.37 $\pm$ 4.48	90.80 $\pm$ 2.31	0.85 $\pm$ 0.04
	RWS	71.88 $\pm$ 4.66	75.70 $\pm$ 5.12	0.61 $\pm$ 0.08
	PSG+RWS	86.09 $\pm$ 2.67	89.86 $\pm$ 2.01	0.83 $\pm$ 0.03

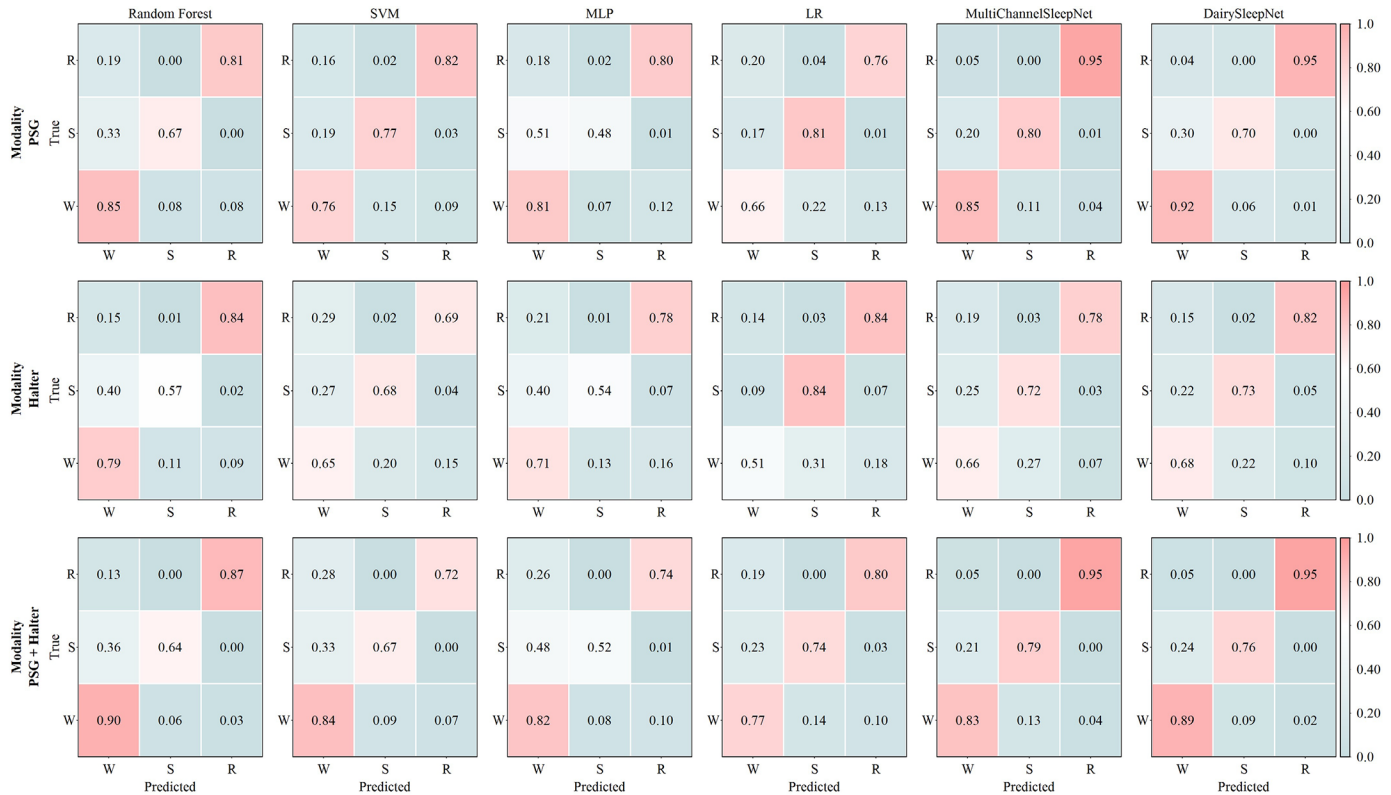


Figure 5. Normalized confusion matrices of DairySleepNet and the 5 baseline models (RF, SVM, MLP, LR, and MultiChannelSleepNet) under the 3 input modalities (PSG, RWS, and PSG+RWS), providing a per-class view of classification patterns. W, Wake; S, Sleep; R, Rumination.

ing uninformative segments. The linear complexity of the recurrence is advantageous in the fusion encoder, where the effective sequence length scales with the number of channels. The sensitivity analysis further showed that DairySleepNet reached strong performance with a compact configuration, which suggests that its advantage derives from more efficient per-layer use of temporal information rather than from greater depth.

Hypnogram Analysis

Figure 6 presents the hypnogram by ground truth annotation and the corresponding predictions from DairySleepNet with PSG input for a randomly selected animal (Cow 7, Fold 4) over a full nocturnal recording period. Segments labeled as M denote periods of missing data that were excluded from both training and evaluation.

The predicted hypnogram closely reproduced the overall temporal structure of the ground truth. The model correctly captured the characteristic polyphasic sleep pattern of dairy cows, in which sleep and rumination bouts alternate cyclically throughout the night rather than consolidating into a single prolonged period as in humans (Ternman et al., 2019). Large-scale behavioral

transitions were identified with high fidelity, including the extended rumination episodes in the early and late portions of the recording, the consolidated sleep bouts in the first part of the night, and the recurring sleep-rumination cycles during the middle of the night.

At a finer temporal resolution, the model demonstrated reliable detection of sustained sleep and rumination bouts but exhibited difficulty with short-duration and fragmented sleep episodes. For instance, the brief and rapidly alternating sleep-wake transitions observed in the early morning were partially captured in the predictions, though some individual epochs within these fragmented bouts were misclassified as Wake. This pattern is consistent with the confusion analysis presented above, where most Sleep misclassifications involved Wake rather than Rumination. The boundary epochs at state transitions represent a particularly challenging scenario, as the physiological signals at these time points may contain mixed characteristics of both the preceding and the succeeding state. Despite these localized errors, the predicted hypnogram preserved the overall temporal distribution and bout structure of each behavioral state, suggesting that DairySleepNet captured the dominant spectral patterns associated with each class even when individual epoch-level predictions were occasionally incorrect.

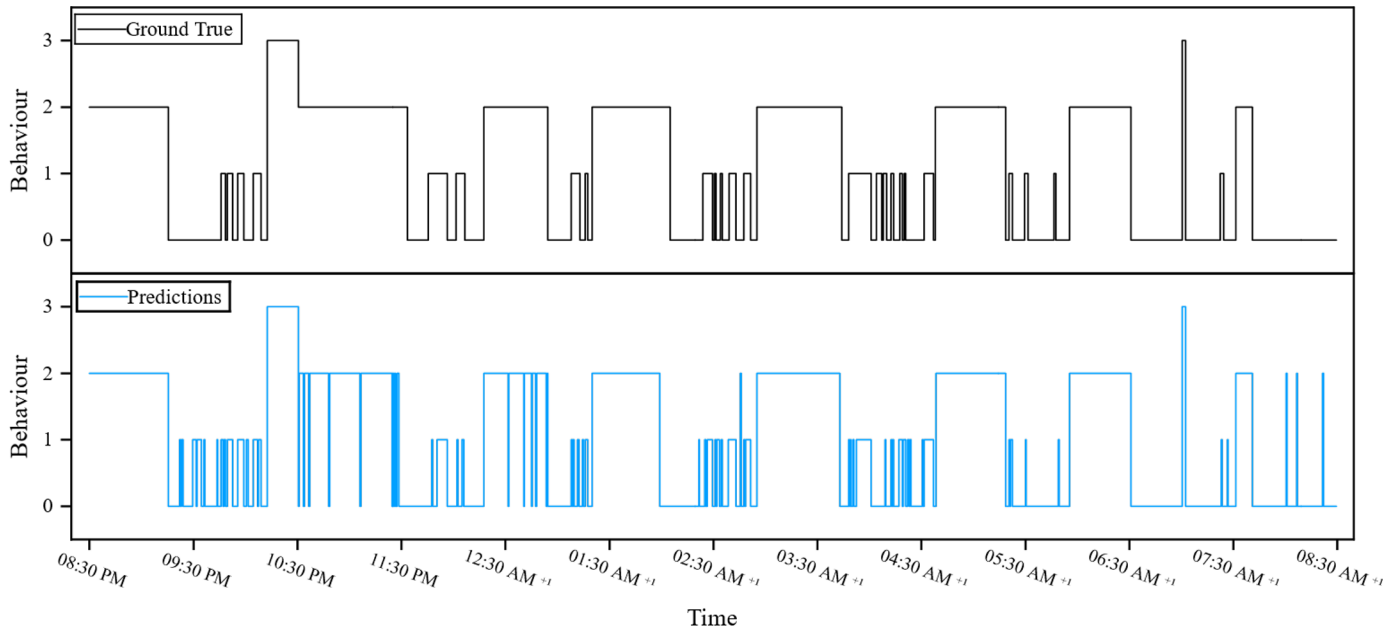


Figure 6. Hypnogram comparison between ground truth and DairySleepNet predictions (PSG input) for Cow 7 (Fold 4) over a full nocturnal recording (8:30 p.m. to 8:30 a.m.). Behavioral states are encoded on the y-axis as Wake (0), Sleep (1), Rumination (2), and Missing (3). Missing segments denote periods of missing data excluded from analysis.

The fragmented errors observed here also point to a limitation of the current design, in which each 30-s epoch is classified independently without reference to neighboring epochs. The hypnogram shows patterns that resemble the cyclic sleep transitions reported in other mammals, although whether dairy cattle follow similarly structured transitions remains to be confirmed. If such regularities exist, sequence-to-sequence architectures that jointly classify several consecutive epochs could exploit the temporal dependence between states and reduce the fragmented misclassifications seen in the early morning period.

Sensitivity Analysis

To examine the effect of model depth on classification performance, we evaluated DairySleepNet under 9 configurations by varying the number of Mamba Blocks in the single-channel encoders N_S and in the multichannel fusion encoder N_M . All experiments were conducted using PSG input with the remaining hyperparameters held constant.

As shown in Table 4, the overall performance was robust to the choice of N_S and N_M , with accuracy ranging from 88.64% to 90.80%, macro F1 from 84.43% to 86.37%, and kappa from 0.81 to 0.85 across all 9 configurations. The total variation in macro F1 was less than 2 percentage points, indicating that no single configuration led to a substantial degradation in performance.

Among the configurations tested, configuration $N_S = 4$ and $N_M = 4$ achieved the best performance on both accuracy (90.80%) and macro F1 (86.37%) and was adopted as the default configuration for all experiments reported. Several configurations yielded closely comparable results, notably $N_S = 16$, $N_M = 8$ (macro F1 = 86.30%) and $N_S = 8$, $N_M = 2$ (macro F1 = 86.09%), with the latter exhibiting the lowest cross-fold variability (accuracy standard deviation of 2.23%).

Examining the effect of each parameter separately revealed that N_M had a somewhat larger influence than N_S when the single-channel encoder is shallow. With $N_S = 4$, varying N_M from 2 to 8 produced a macro F1 spread of 1.94 percentage points (84.43% to 86.37%), whereas with $N_S = 8$, the same variation in N_M yielded a spread of only 0.74 percentage points (85.35% to 86.09%). This pattern suggests that a shallow single-channel encoder places greater reliance on the fusion stage to compensate for less refined per-channel representations, while a deeper single-channel encoder produces features that are already sufficiently discriminative, reducing the sensitivity to the fusion depth. Increasing N_S beyond 8 did not yield further improvements, as the $N_S = 16$ configurations showed no consistent advantage over $N_S = 8$, suggesting that additional layers do not contribute meaningful representational capacity for the 29-step temporal sequences used in this study.

Several considerations nevertheless bound the scope of these findings. Although the model proved robust to

the choice of architectural depth, its evaluation rested on a limited data set. The recordings comprised 7 Holstein cows from a single tie-stall facility, which constrains generalization to larger populations, other breeds, and alternative housing systems. The annotations were produced by a single trained scorer, so inter-rater reliability could not be assessed (Rosenberg and Van Hout, 2013), and some epoch-level disagreements between the model and the reference may reflect annotation ambiguity rather than model error.

CONCLUSION

This study presented DairySleepNet, a Mamba-based deep learning framework for automated sleep behavior classification in dairy cows using multi-channel physiological signals. By combining time-frequency representations with a 2-stage encoding architecture that captures both within-channel temporal features and cross-channel interactions, the proposed framework enabled reliable discrimination among wakefulness, sleep, and rumination under a rigorous animal-level evaluation protocol. Our findings highlighted the value of neurophysiological signals for sleep characterization in cattle while also revealing the current limitations of accelerometer-based sensors when used in isolation. Beyond establishing the feasibility of automated sleep classification in dairy cattle, this work opened promising directions for finer-grained sleep staging, sequence-level temporal modeling, and the integration of complementary non-invasive sensing modalities. We envision that such advances will ultimately support scalable, welfare-oriented sleep monitoring in commercial dairy farming and contribute to a deeper understanding of sleep biology in ruminants.

NOTE

All experimental procedures involving animals were approved by the Cantonal Veterinary Office of Zürich, Switzerland (approval number: ZH042/2023), and were

conducted in accordance with Swiss legislation on animal protection and welfare. This study received no external funding, and data is available from the corresponding author upon request.

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REFERENCES

- Aboalayon, K. A. I., M. Faezipour, W. S. Almuhammadi, and S. Moslehpour. 2016. Sleep stage classification using EEG signal analysis: A comprehensive survey and new investigation. *Entropy (Basel)* 18:272. <https://doi.org/10.3390/e18090272>.
- Allen, J. B., and L. R. Rabiner. 1977. A unified approach to short-time Fourier analysis and synthesis. *Proc. IEEE* 65:1558–1564. <https://doi.org/10.1109/PROC.1977.10770>.
- Ba, J. L., J. R. Kiro, and G. E. Hinton. 2016. Layer normalization. *arXiv preprint arXiv:1607.06450*. <https://doi.org/https://doi.org/10.48550/arXiv.1607.06450>.
- Breiman, L. 2001. Random forests. *Mach. Learn.* 45:5–32. <https://doi.org/10.1023/A:1010933404324>.
- Chambon, S., M. N. Galtier, P. J. Arnal, G. Wainrib, and A. Gramfort. 2018. A deep learning architecture for temporal sleep stage classification using multivariate and multimodal time series. *IEEE Trans. Neural Syst. Rehabil. Eng.* 26:758–769. <https://doi.org/10.1109/TNSRE.2018.2813138>.
- Cirelli, C., and G. Tononi. 2008. Is sleep essential? *PLoS Biol.* 6:e216. <https://doi.org/10.1371/journal.pbio.0060216>.
- Cohen, J. 1960. A coefficient of agreement for nominal scales. *Educ. Psychol. Meas.* 20:37–46. <https://doi.org/10.1177/001316446002000104>.
- Cook, N. B. 2020. Symposium review: The impact of management and facilities on cow culling rates. *J. Dairy Sci.* 103:3846–3855. <https://doi.org/10.3168/jds.2019-17140>.
- Cortes, C., and V. Vapnik. 1995. Support-vector networks. *Mach. Learn.* 20:273–297. <https://doi.org/10.1023/A:1022627411411>.
- Dai, Y., X. Le, G. Wu, B. Wang, and G. Yan. 2023. MultiChannelSleepNet: A transformer-based model for automatic sleep stage classification with PSG. *IEEE J. Biomed. Health Inform.* 27:4204–4215. <https://doi.org/10.1109/JBHI.2023.3284160>.
- Eldele, E., Z. Chen, C. Liu, M. Tan, J. G. Huang, and X. Li. 2021. An attention-based deep learning approach for sleep stage classification with single-channel EEG. *IEEE Trans. Neural Syst. Rehabil. Eng.* 29:809–818. <https://doi.org/10.1109/TNSRE.2021.3076234>.
- Goldberger, A. L., L. A. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C. K. Peng, and H. E. Stanley. 2000. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* 101:E215–E220. <https://doi.org/10.1161/01.CIR.101.23.e215>.
- Gu, A., and T. Dao. 2023. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*. <https://doi.org/https://doi.org/10.48550/arXiv.2312.00752>.
- Guillodo, E., C. Lemey, M. Simonnet, M. Walter, E. Baca-García, V. Masetti, S. Moga, M. Larsen, R. Ropars, and S. Berrouguet. 2020. Clinical applications of mobile health wearable-based sleep monitoring: Systematic review. *JMIR Mhealth Uhealth* 8:e10733. <https://doi.org/10.2196/10733>.
- Guo, M., X. Han, H. Liu, J. Zhu, J. Zhang, Y. Bai, and G. Ni. 2025. MI-Mamba: A hybrid motor imagery electroencephalograph classification model with Mamba's global scanning. *Ann. N. Y. Acad. Sci.* 1544:242–253. <https://doi.org/10.1111/nyas.15288>.

Table 4. Sensitivity analysis of DairySleepNet performance under varying numbers of Mamba blocks in the single-channel encoder N_s and the multichannel fusion encoder N_m , evaluated with PSG input

N_s	N_m	ACC	Macro F1	κ
4	2	88.64 ± 2.81	84.43 ± 4.12	0.81 ± 0.05
4	4	90.80 ± 2.31	86.37 ± 4.48	0.85 ± 0.04
4	8	89.58 ± 3.06	85.28 ± 4.81	0.83 ± 0.05
8	2	90.34 ± 2.23	86.09 ± 3.96	0.84 ± 0.04
8	4	90.23 ± 2.39	85.35 ± 4.24	0.84 ± 0.04
8	8	89.94 ± 2.72	85.42 ± 4.29	0.83 ± 0.05
16	2	89.65 ± 2.21	85.22 ± 3.53	0.83 ± 0.04
16	4	88.81 ± 3.03	84.68 ± 3.99	0.81 ± 0.05
16	8	90.29 ± 2.69	86.30 ± 4.06	0.84 ± 0.05

- He, K., X. Zhang, S. Ren, and J. Sun. 2016. Deep residual learning for image recognition. *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* 770–778. <https://doi.org/10.1109/CVPR.2016.90>.
- Hendrycks, D., and K. Gimpel. 2016. Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415*. <https://doi.org/https://doi.org/10.48550/arXiv.1606.08415>.
- Hunter, L. B., A. Baten, M. J. Haskell, F. M. Langford, C. O'Connor, J. R. Webster, and K. Stafford. 2021b. Machine learning prediction of sleep stages in dairy cows from heart rate and muscle activity measures. *Sci. Rep.* 11:10938. <https://doi.org/10.1038/s41598-021-90416-y>.
- Kemp, B., A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Obery. 2000. Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG. *IEEE Trans. Biomed. Eng.* 47:1185–1194. <https://doi.org/10.1109/10.867928>.
- Korkalainen, H., T. Leppänen, B. Duce, S. Kainulainen, J. Hasan, and J. Töyräs. 2021. Detailed assessment of sleep architecture with deep learning and shorter epoch-to-epoch duration reveals sleep fragmentation of patients with obstructive sleep apnea. *IEEE J. Biomed. Health Inform.* 25:2567–2574. <https://doi.org/10.1109/JBHI.2020.3043507>.
- Krueger, J. M., D. M. Rector, S. Roy, H. P. A. Van Dongen, G. Belenky, and J. Panksepp. 2008. Sleep as a fundamental property of neuronal assemblies. *Nat. Rev. Neurosci.* 9:910–919. <https://doi.org/10.1038/nrn2521>.
- Kull, J. A., K. L. Proudfoot, G. M. Pighetti, J. M. Bewley, B. F. O'Hara, K. D. Donohue, and P. D. Krawczel. 2019. Effects of acute lying and sleep deprivation on the behavior of lactating dairy cows. *PLoS One* 14:e0212823. <https://doi.org/10.1371/journal.pone.0212823>.
- Lee, C.-H., H. Kim, B. C. Yoon, and D.-J. Kim. 2025. Toward foundational model for sleep analysis using a multimodal hybrid-self-supervised learning framework. *IEEE Trans. Cybern.* 55:5619–5632. <https://doi.org/10.1109/TCYB.2025.3603608>.
- Lewis, D. D., R. E. Schapire, J. P. Callan, and R. Papka. 1996. Training algorithms for linear text classifiers. *Proc. 19th Annu. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr.* 298–306. <https://doi.org/10.1145/243199.243277>.
- Li, X., S. Yan, Y. Wu, C. Dai, and Y. Guo. 2025. A novel state space model with dynamic graphic neural network for EEG event detection. *Int. J. Neural Syst.* 35:2550008. <https://doi.org/10.1142/S012906572550008X>.
- Loshchilov, I., and F. Hutter. 2019. Decoupled weight decay regularization. *Proc. 7th Int. Conf. Learn. Represent. (ICLR)*. <https://doi.org/10.48550/arXiv.1711.05101>.
- Palotti, J., R. Mall, M. Aupetit, M. Rueschman, M. Singh, A. Sathyanarayana, S. Taheri, and L. Fernandez-Luque. 2019. Benchmark on a large cohort for sleep-wake classification with machine learning techniques. *npj. NPJ Digit. Med.* 2:50. <https://doi.org/10.1038/s41746-019-0126-9>.
- Quan, S. F., B. V. Howard, C. Iber, J. P. Kiley, F. J. Nieto, G. T. O'Connor, D. M. Rapoport, S. Redline, J. Robbins, J. M. Samet, and P. W. Wahl. 1997. The Sleep Heart Health Study: design, rationale, and methods. *Sleep* 20:1077–1085. <https://doi.org/10.1093/sleep/20.12.1077>.
- Rechtschaffen, A., and A. Kales, eds. 1968. A manual of standardized terminology, techniques and scoring system for sleep stages of human subjects. *Brain Information Service/Brain Research Institute*, University of California, Los Angeles.
- Riaboff, L., S. Poggi, A. Madouasse, S. Couvreur, S. Aubin, N. Bédère, E. Goumand, A. Chauvin, and G. Plantier. 2020. Development of a methodological framework for a robust prediction of the main behaviours of dairy cows using a combination of machine learning algorithms on accelerometer data. *Comput. Electron. Agric.* 169:105179. <https://doi.org/10.1016/j.compag.2019.105179>.
- Riaboff, L., L. Shalloo, A. F. Smeaton, S. Couvreur, A. Madouasse, and M. T. Keane. 2022. Predicting livestock behaviour using accelerometers: A systematic review of processing techniques for ruminant behaviour prediction from raw accelerometer data. *Comput. Electron. Agric.* 192:106610. <https://doi.org/10.1016/j.compag.2021.106610>.
- Rombach, M., A. Münger, J. Niederhauser, K.-H. Südekum, and F. Schori. 2018. Evaluation and validation of an automatic jaw movement recorder (RumiWatch) for ingestive and rumination behaviors of dairy cows during grazing and supplementation. *J. Dairy Sci.* 101:2463–2475. <https://doi.org/10.3168/jds.2016-12305>.
- Rosenberg, R. S., and S. Van Hout. 2013. The American Academy of Sleep Medicine inter-scoring reliability program: sleep stage scoring. *J. Clin. Sleep Med.* 9:81–87. <https://doi.org/10.5664/jcsm.2350>.
- Rumelhart, D. E., G. E. Hinton, and R. J. Williams. 1986. Learning representations by back-propagating errors. *Nature* 323:533–536. <https://doi.org/10.1038/323533a0>.
- Shannon, C. E. 1949. Communication in the presence of noise. *Proc. IRE* 37:10–21. <https://doi.org/10.1109/JRPROC.1949.232969>.
- Srivastava, N., G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. 2014. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15:1929–1958.
- Suarez-Trujillo, A., N. Hoang, L. Robinson, C. J. McCabe, D. Conklin, R. C. Minor, J. Townsend, K. Plaut, U. Z. George, J. Boerman, and T. M. Casey. 2022. Effect of circadian system disruption on the concentration and daily oscillations of cortisol, progesterone, melatonin, serotonin, growth hormone, and core body temperature in periparturient dairy cattle. *J. Dairy Sci.* 105:2651–2668. <https://doi.org/10.3168/jds.2021-20691>.
- Supratak, A., H. Dong, C. Wu, and Y. Guo. 2017. DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG. *IEEE Trans. Neural Syst. Rehabil. Eng.* 25:1998–2008. <https://doi.org/10.1109/TNSRE.2017.2721116>.
- Ternman, A., L. Hänninen, M. Pastell, S. Agenes, and P. P. Nielsen. 2012. Sleep in dairy cows recorded with a non-invasive EEG technique. *Appl. Anim. Behav. Sci.* 140:25–32. <https://doi.org/10.1016/j.applanim.2012.05.005>.
- Ternman, A., E. Nilsson, P. P. Nielsen, M. Pastell, L. Hänninen, and S. Agenes. 2019. Rapid eye movement sleep time in dairy cows changes during the lactation cycle. *J. Dairy Sci.* 102:5458–5465. <https://doi.org/10.3168/jds.2018-15950>.
- Ternman, A., M. Pastell, S. Agenes, C. Strasser, C. Winckler, P. P. Nielsen, and L. Hänninen. 2014. Agreement between different sleep states and behaviour indicators in dairy cows. *Appl. Anim. Behav. Sci.* 160:12–18. <https://doi.org/10.1016/j.applanim.2014.08.014>.
- Vaswani, A., N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. 2017. Attention is all you need. *Adv. Neural Inf. Process. Syst.* 30:5998–6008. <https://doi.org/10.48550/arXiv.1706.03762>.
- Walch, O., Y. Huang, D. Forger, and C. Goldstein. 2019. Sleep stage prediction with raw acceleration and photoplethysmography heart rate data derived from a consumer wearable device. *Sleep* 42:zs180. <https://doi.org/10.1093/sleep/zsz180>.
- Zhang, G. Q., L. Cui, R. Mueller, S. Tao, M. Kim, M. Rueschman, S. Mariani, D. Mobley, and S. Redline. 2018. The National Sleep Research Resource: towards a sleep data commons. *J. Am. Med. Assoc.* 320:1351–1358. <https://doi.org/10.1093/jamia/ocy064>.
- Zhou, D., Q. Xu, J. Wang, H. Xu, L. Kettunen, Z. Chang, and F. Cong. 2022. Alleviating class imbalance problem in automatic sleep stage classification. *IEEE Trans. Instrum. Meas.* 71:4006612. <https://doi.org/10.1109/TIM.2022.3191710>.

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